

SWADHIN AGRAWAL^{id}

PhD (Senior Research Fellow) @Dept. EECS, IISER'B

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Education

IISER Bhopal

PhD in Robotics,

Supervisor: Prof. Sujit P.B,

Co-supervisor: Dr Andreagiovanni Reina

8.67 CPI

Aug. 2020 – Present

Bhopal, Madhya Pradesh

IISER Bhopal

BS-MS,

Physics (major), EECS (minor),

Supervisor: Dr Deepak Kumar

8.54 CPI

Aug. 2015 – May 2020

Bhopal, Madhya Pradesh

Gangadhar Meher Junior College

Higher secondary school

84.5 %

May. 2013 – May 2015

Sambalpur, Odisha

Relevant Coursework

- Control Systems
- Theory of Computation
- Intelligent robotics
- Model verification
- Introduction to ML
- Network Science
- Computational Geometry
- Linear Algebra
- Math Methods I & II
- Multivariable Calculus
- Probability and Statistics
- Behavioral Biology
- Self-Driving Cars
- SLAM

Technical Skills

Languages: Python, C, C++, C#, MATLAB

Markup Languages: HTML, Latex, XML

Script Languages: Bash, Expect, Docker

Developer Tools: VS Code, Blender, OpenFoam

Frameworks: Linux, GitHub, ROS 1, ROS 2, Docker Containers

Simulators: Unity, Gazebo, Webots

Python libraries: Statmodels, Pandas, Numpy, Matplotlib, Multiprocessing, Numba, Ray

Robotic Tools: SITL, Ardupilot, PX4, Mission Planner, Q-Ground Control

Teaching Activities

IISER Bhopal

Teaching Assistant,

Competitive Programming ECS342

Jan 2025 – Apr 2025

Bhopal, MP

IISER Bhopal

Teaching Assistant,

Control systems ECS323

Aug 2023 – Dec 2023

Bhopal, MP

IISER Bhopal

Teaching Assistant,

Introduction to programming ECS102

Jan 2023 – May 2023

Bhopal, MP

IISER Bhopal

Teaching Assistant,

Computer Organization ECS409

Aug 2022 – Dec 2022

Bhopal, MP

IISER Bhopal

Teaching Assistant,

Data structures and algorithms ECS202

Jan 2021 – May 2022

Bhopal, MP

IISER Bhopal

Teaching Assistant,
Control systems ECS323

Aug 2021 – Nov 2021

Bhopal, MP

IISER Bhopal

Teaching Assistant,
Introduction to programming ECS102

Jan 2021 – May 2021

Bhopal, MP

IISER Bhopal

Teaching Assistant,
Control systems ECS323

Aug 2020 – Nov 2020

Bhopal, MP

Academic Activities

University of Konstanz, Max-Planck Institute of Animal Behaviour

ZuKonnnect Fellow under *Dr Liang Li*,
Exploring collective motion through evolutionary game theory

Aug 2024 – Dec 2024

Konstanz, Germany

The project aims to explore collective behavior through the lens of evolutionary game theory. Most contemporary models for collective behavior assume deterministic individual movements under noisy conditions. However, individuals, in reality, can be irrational and may refuse to follow deterministic rules. Our primary goal is to investigate how these irrational individuals (defectors) affect collective behavior, considering both the pros and cons. We will explore four sub-objectives:

1. The impact of defectors on collective behavior.
2. Dynamic role switching and the emergence of leadership in swarms.
3. Analyzing the influence of varying time scales on individual behaviors within the game framework.

University of Konstanz, Max-Planck Institute of Animal Behaviour

Visiting Researcher under *Dr Liang Li*,
Combining evolutionary game theory to collective behavior

Jan 2024 – March 2024

Konstanz, Germany

Understanding the fundamental mechanisms underlying collective behavior is pivotal in the study of collective behavior. Classical models, such as the Vicsek or Couzin models, often neglect psychological influences, operating on the assumption that each individual strictly adheres to predefined rules. These models then add noise to account for the variations observed in collective formations. However, in the natural world, agents—be they animals or humans—do not always act rationally or in line with these rules. There are instances where individuals might disregard a rule, even if adherence would yield optimal collective outcomes. In this study, we employ evolutionary game theory to investigate how the psychology of individuals impacts collective intelligence. Specifically, we simplify psychological effects into two categories: collaborators, who follow the rules, and defectors, who do not. Our aim is to determine if the presence of defectors can, in certain contexts, benefit the collective.

IISER Bhopal

MS Thesis,
Effect of Gaussian curvature on frictional interaction between an ultra-thin film and a soft-substrate

May 2019 – May 2020

Bhopal, MP

- * In this project, we explored the role of geometry in modifying the strength of adhesion and hence friction between a few nano-meter thin polystyrene film and a soft hydrogel substrate using RGB image processing techniques on acquired data through long timed imaging.
- * In our experiments, we loaded the ultra-thin films on top of the swollen hydrogel surfaces that shrink over time due to the homogeneous evaporation of water.
- * By measuring the compression of the film and the shrinking rate of the surface through image analysis using MATLAB, we found that the frictional interaction between the substrate and the film depends on the underlying Gaussian curvature of the substrate.
- * As part of the thesis, I also designed the automated experimental setup using Laser and (Olympus & Nikon) DSLR camera, both controlled through Arduino UNO.

Robert Bosch Center for Cyber-Physical Systems

Intern under *Dr Raghu Krishnapuram*,
Depth estimation from Monocular image sequences using machine learning methods

May 2019 – July 2019

Bangalore, Karnataka

- * This project was a part of the MBZIRC 2020 competition. In this project, our target was to use a monocular camera and estimate the depth of the objects in the scene.

- We explored various kinds of existing methods like using Convolutional Neural Networks, Geometrical approaches and Dense Nets.
- We tried to reinvent Struct2Depth by Google Brain, DEEPTAM and Demon by LMB-Freiburg, SFMLearner by Tinghui Zhou et al., Disparity learning due to time (in SFMLearner) as well as Space Disparity Learning (MonoDepth by Godard et al.).

IISER Bhopal

June 2018 – July 2018

Intern under [Dr Kushal Kumar Shah](#),

Bhopal, MP

[On finding the exact Analytical solution of Fokker Planck Equation for Plasma](#)

- In this project, we made an attempt to find the exact solution for the Fokker Planck Equation derived for a Paul trap which was a higher-order & higher degree multivariable partial differential equation.
- This equation gives rise to Riccati Equations, which we did rethinking and made an attempt to solve numerically in Python in order to understand the possibilities of finding the exact solutions. Results of this project can be found in the website link provided above.
- As a part of the internship, we also presented poster of our work at the Plasma conference held at University of Delhi, in 2018. Our work has also been acknowledge as the preliminary work which was published in [AIP Physics of Plasma](#).

Poornaprajna Institute of Scientific Research

May 2017 – July 2017

Intern under [Dr R Srikant](#),

Bangalore, Karnataka

[On Approximating Universal Quantum Computation](#)

- Here, I learned the basic aspects of Qubits and Quantum Computation, Quantum Gates and Circuits.
- Also, I used the IBM Quantum Gates and created Quantum gates in Mathematica using a software package called QETLAB.
- I explored Quantum Computation and Quantum Information from McMahon and Nielsen and Chuang's books.

Challenges

[MBZIRC 2023](#) | *Semi-Finalist Team LUNA , Student Leader*

Dec 2021 - Aug 2022

- Implemented the decentralized swarm motion algorithm for inspection and payload lifting from the target vessel in ROS2, Gazebo Ignition MBZIRC environment and Docker.
- Implemented pick and place task for the Robotic arm.
- Implemented lawn-mower algorithm for target vessel search on a delta-fixed-wing vehicle.
- Implemented PID controlled go-to-goal algorithm for unmanned surface vehicle.

[VORC 2020](#) | *Team Unmet dependencies, Student Leader*

Nov 2020 - Dec 2020

- Participated in the competition to get exposure to the latest software frameworks like Docker and ROS1.

Scientific Reviews

[ETA AV 2025](#) | *Conference*

2025

Journal of Aerospace Engineering 2023 | *Journal*

2023

[COCOON 2022](#) | *Conference*

2022

[Swarm Intelligence](#) | *Special Issue, Springer*

June 2021

Publications

Conferences

1. Swadhin Agrawal, Sujit P. Baliyarasimhuni, and Andreagiovanni Reina. Response threshold distributions to improve best-of-n decisions in minimalistic robot swarms. In *Swarm Intelligence*, pages 350–359, Cham, 2022
2. Swadhin Agrawal, Jitesh Jhawar, Andreagiovanni Reina, Sujit P. Baliyarasimhuni, Heiko Hamann, and Liang Li. Impact of individual defection on collective motion. In *Swarm Intelligence*, pages 127–140, Cham, 2024

To be submitted

1. Swadhin Agrawal, Sujit P. Baliyarasimhuni, and Andreagiovanni Reina. Adaptive decentralized collective best-of-n decision-making for minimalistic robot swarms
2. Swadhin Agrawal, Sujoy Bhore, Joseph S.B. Mitchell, and Sujit P. Baliyarasimhuni. Multi-robot searching with limited sensing range for static and mobile intruders
3. Swadhin Agrawal, Manav Mishra, Sujoy Bhore, and Sujit P. Baliyarasimhuni. Leader-based herding of robot swarm amidst obstacles

Poster Presentation

ANTS conference <i>participant</i>	2024
WIVACE conference <i>participant</i>	2024
ANTS conference <i>participant</i>	2022

Fellowships

Fellowship for Training of Young Scientist <i>40th M.P Young Scientist Congress</i>	Feb 2025 - Feb 2027 <i>Vikram University</i>
KSCB Travel Grant <i>Germany</i>	July 2024 <i>University of Konstanz</i>
Senior Research Fellowship <i>Institute</i>	Aug 2022 – Present <i>IISER'B</i>
FAAC Travel Grant <i>For attending ANTS 2022 Conference</i>	Nov 2022 <i>IISER'B</i>
Junior Research Fellowship <i>Institute</i>	Aug 2020 – Aug 2022 <i>IISER'B</i>
Innovation in Science Pursuit for Inspired Research Fellowship <i>DST-Inspire</i>	Aug 2015 – May 2020 <i>IISER'B</i>

Societies & Memberships

IEEE Robotics & Automation Society (RAS) <i>Student Member</i>	2020 - Present
Society for the Study of Evolution (SSE) <i>Student Member</i>	2020 - Present

Invited talks

Center for Advanced Studies and Collective Behavior Seminar <i>Impact of Individual Defection on Collective Motion</i>	2024 <i>University of Konstanz</i>
Hegau-Bodensee Seminar <i>What is Swarm Robotics?</i>	2024 <i>University of Konstanz</i>
Summer Outreach Program <i>Robotics in Lab and Field</i>	2022 <i>IISER'B</i>
American India Foundation <i>Role Model Interaction</i>	Dec 2022 <i>IISER'B</i>

Leadership

Team Leader

MBZIRC 2023

2021-2022

Team LUNA, IISER'B

Founder & Coordinator

Rocket & Satellite Designing Club

2018-2020

IISER'B

Secretary

Sports Council

Nov 2017 - May 2018

IISER'B

Coordinator

Computer Science & Programming Club

2016

IISER'B

Awards

40th M.P Young Scientist

Vikram University, Ujjain

2025

Madhya Pradesh

10th Bhopal Vigyan Mela

2nd Vigyan Pavillion Prize, Research category

2023

Bhopal

Student of the Year

Principal

2015

Gangadhar Meher Junior College

Workshops & Summer/Winter schools

Konstanz School of Collective Behaviour

Participant

July 2024

University of Konstanz, Germany

Summer school on Multi-Robot Systems

Participant

Aug 2022

Czech Technical University, Prague

Winter school on SLAM in deformable environments

Participant (2nd prize on challenge)

July 2021

University of Sydney

Extracurricular & FunFacts

India International Science Fest 2022

Participant

Jan 2023

MNIT Bhopal, India

Hobbies

- Playing Guitar
- Gardening
- Kabaddi
- Pistol Shooting
- Martial Arts
- Basketball
- Photography